

# Methods: Arp\_220

ALMA Thesis Planner (automated pipeline)

## 0.1 0. Framing and conventions to adopt before you touch the data

You are making one measurement, repeated across a bounded set of positions and velocities: the **line-to-continuum ratio**  $R(v) = I_{\text{line}}(v)/I_{\text{cont,local}}$  of  $^{12}\text{CO}$  J=4–3 across the two nuclei of Arp 220, together with a full characterization of the 650  $\mu\text{m}$  continuum against which that ratio is defined. Everything below serves that measurement.

Fix three conventions now and hold them for the whole thesis, because they will keep you from writing claims the data cannot support:

1. **Never write  $\tau$ .** Write *apparent absorption depth*  $D(v) \equiv -\ln R(v)$ , or *line-to-continuum ratio*, and say in the text at first use that  $D$  is not an optical depth. In these nuclei the 650  $\mu\text{m}$  dust and the CO are co-spatial and the dust is optically thick, so a "continuum" measurement is a nearly opaque surface at unknown depth, and CO emission from the same gas parcel fills in absorption along the identical sightline. Every  $D$  you quote is a lower bound.
2. **The sign of ( $I_{\text{line}} - I_{\text{cont}}$ ) is a result, not an input.** The "deep CO absorption" in the proposal abstract refers to CO 6–5 and 3–2 in other member OUS that are not in your dataset. Whether CO(4–3) goes below the continuum here is something you will determine.
3. **Interpret in brightness temperature, not geometry.** A line appears in absorption when the continuum brightness temperature  $T_{\text{B,cont}}$  exceeds the CO(4–3) excitation temperature  $T_{\text{ex}}$  along the same gas, and in emission when it does not. Do not use the foreground-screen language ("redshifted absorption" "gas falling in"); the geometry does not license it.

Set up a git repository on day one containing your CASA scripts, Python analysis notebooks, and a running lab notebook in Markdown. Every number that ends up in the thesis should be traceable to a script in that repository. Record the CASA version string in every log.

## 0.2 1. Weeks 1–2 — Retrieval, spectral-setup verification, and the QA2-first decision

**Retrieval.** Download member OUS 'uid://A001/X1289/X1e6' of project 2017.1.00735.S from the ALMA Science Archive (<https://almascience.nrao.edu/aq/>), searching either by project code or by the OUS UID. Take the full delivery: the pipeline-calibrated measurement sets, the QA2 image products (cube and continuum), the weblog, and the calibration/imaging scripts. Untar into a read-only 'raw/' tree and never edit anything inside it; all work happens in a separate 'work/' tree.

**Read the weblog before the data.** The QA2 weblog tells you the achieved beam, the achieved RMS, which spectral windows were flagged, what the flux calibrator was and what model was used, and whether the pipeline did any continuum subtraction. Extract and tabulate: achieved synthesized beam (major, minor, PA) for each product, achieved continuum RMS, achieved per-channel line RMS, total on-source time, and the number of antennas and baseline range. The archive

tabulates 0.086'' angular resolution, 0.746 km/s native velocity resolution, 302 s integration, line sensitivity 8.336 mJy/beam at 10 km/s, and continuum sensitivity 0.2723 mJy/beam — treat all of these as *predicted/tabulated* values and replace each with the value you measure from the delivered images. Your error budget uses your measured numbers, not the archive's.

**Verify the spectral setup from the MS header, not from the annotation.** Run ‘listobs’ on each measurement set and, in the CASA Python environment, use ‘msmd’ to pull the exact spectral-window edges, channel widths, number of channels, and the frame in which frequencies are recorded (TOPO vs LSRK — this matters). Confirm the four windows: 451.9–453.8, 453.6–455.6, 463.8–465.8, 465.6–467.6 GHz.

Then do the line identification yourself. The archive's line list is annotated as generated from general knowledge and flagged for verification against Splatalogue, so verify it. Query Splatalogue for  $^{12}\text{CO}$  J=4–3 and confirm the rest frequency 461.041 GHz. Apply the systemic redshift that the annotation *assumes*,  $z = 0.018126$  ( $cz = 5434$  km/s), and confirm for yourself that it lands at  $\approx 452.8$  GHz — this places the line in the 451.9–453.8 and 453.6–455.6 GHz pair, whose combined  $\sim 3.7$  GHz spans roughly 2450 km/s. Carry the assumption explicitly: in the thesis, write “ $\approx 452.8$  GHz, assuming  $z = 0.018126$ ,” the first several times, and state that the sky-frequency placement was confirmed against the MS header. Cross-check by converting the spw edges to LSRK velocity relative to CO(4–3) at that redshift and writing down the actual velocity range covered, in km/s relative to  $cz = 5434$  km/s. That velocity axis is the one you will use everywhere.

Do the same Splatalogue sweep over the 463.8–465.8 and 465.6–467.6 GHz windows. The annotation offers high-J  $\text{HC}_3\text{N}$  (J=51–50, rest  $\approx 464.0$  GHz; J=52–51, rest  $\approx 473.1$  GHz) as *tentative* candidates and is itself agnostic as to whether such features would be redshifted or near-rest-frequency serendipitous. That identification is **not established**, and you are not going to establish it. Treat both of those windows as **additional continuum**. If a feature appears there, report it as a tentative, clearly flagged secondary result with the candidate identification stated as a candidate and the frame ambiguity stated alongside it — one figure and one paragraph, not a chapter.

**Commit to the QA2-first path.** Your primary analysis path is the delivered QA2 image products. Restoring a Cycle-5 Band 8 pipeline calibration requires a matched CASA version and can silently consume weeks. Therefore:

- All headline measurements — continuum photometry, Gaussian fits, the  $\sigma_{\text{D}}$  map, the sign map, the branch products — are computed from QA2 products first.
- Re-imaging with ‘tclean’ from the calibrated MS is an **enhancement**, attempted only after the QA2-based measurements exist end-to-end.
- Write the abandonment date into your notebook now. If re-imaging has not produced a usable image by the end of week 8, you stop, and the attempt is documented as a result (what you tried, what failed, with the logs) rather than pursued.

Load the QA2 continuum image and cube into CARTA for a first visual look. Confirm you can see two compact nuclei, and record their approximate positions relative to the pointing centre RA, Dec = 233.73843, +23.50322.

### 0.3 2. Week 2–3 — Continuum imaging and nuclear photometry

Work on the QA2 continuum product. If the delivery includes per-spw continuum images, use them; also form a combined continuum from the two line-free windows (463.8–465.8, 465.6–467.6

GHz) for maximum continuum SNR, keeping the effective frequency recorded.

**Photometry and morphology.** Using CASA:

- ‘imstat’ on several signal-free regions to measure the image RMS and confirm it against the weblog value; use the RMS you measure.
- ‘imfit’ with a two-component 2D Gaussian model (one per nucleus) plus a zero-level term, fitted over a box enclosing both nuclei. Record for each nucleus: fitted position with uncertainty, peak intensity (mJy/beam), integrated flux density (mJy), and **deconvolved** major axis, minor axis, and position angle with uncertainties. Note explicitly whether each component is deconvolvable at all — if ‘imfit’ returns a size consistent with the beam, report an upper limit on the size rather than a size.
- Repeat the fit with the fitting box, initial estimates, and background treatment varied, and adopt the spread across those variants as your fitting systematic. Add it in quadrature to the statistical error from ‘imfit’.
- Convert peak intensities to per-beam brightness temperature using the Rayleigh–Jeans relation with your measured beam and the effective frequency, and tabulate  $T_{B,cont}$  for each nucleus. This number is central to the interpretation in §7.

**Flux-scale error budget.** Carry the Band 8 absolute flux-scale uncertainty of roughly 10–20% as a separate, explicitly labelled column in every continuum table. State clearly in the text which quantities it touches — nuclear flux densities, any dust mass you derive, and every brightness-temperature statement — and which it does **not**: R and D are ratios formed within the same spectral setup, so the absolute scale divides out of them. Say this once, plainly, and then keep the two error columns separate throughout.

**No spectral index.** Do not attempt to measure a dust spectral index between the window pairs.  $\Delta\nu/\nu \approx 2.6\%$  across the pairs, so the predicted dust-slope difference is comparable to Band 8 spw-to-spw amplitude systematics and the measurement is uninformative by construction. State this reasoning in one sentence in the methods chapter and spend those days on §3 instead.

**Bounded self-calibration.** With only 302 s on source, self-calibration may or may not be viable, so bound it: run a short ladder of phase-only solution intervals (‘inf’, then a couple of shorter ones) with ‘gaincal’, inspect solution SNR and the fraction of failed solutions in ‘plotms’, and adopt self-cal only if a clear majority of solutions have acceptable SNR and the resulting image RMS improves without altering the fitted nuclear positions or sizes beyond their uncertainties. If no interval qualifies, drop self-cal, keep the diagnostic plots, and document the outcome. Do not iterate past the ladder you defined.

## 0.4 3. Week 3 — The go/no-go tier decision

This is the single most important week of the project, and it ends with a written decision.

**Measure the quantity that actually decides the project: the per-beam continuum brightness.** At  $0.086''$  much of Arp 220’s flux may be resolved out, so the *total* flux is not what matters — the per-beam continuum intensity at each position is what sets your ability to measure R there. Produce a per-beam continuum brightness map (this is simply your calibrated continuum image in mJy/beam, but treat it as the key data product it is).